

DANIEL ROUSSEAU

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Education

University of California San Diego

B.S. in Computer Science and Mathematics

Jul 2025

- **Leadership:** Competitive Programming Club — *Head Mentor & Problem Lead* (2023–2025); led weekly algorithms sessions, organized mock contests, coached ICPC-style teams, and mentored **15+** members for over 2 years
- **Relevant Coursework:** Data Structures & Algorithms, Operating Systems, Databases, Software Engineering, Computer Networks, Distributed Systems, Computer Architecture, Probability & Stochastic Processes

Experience

Software Engineering Intern (ML/AI) *Axented*

Jul 2024 – Oct 2024

- Developed and shipped **3** full-stack Laravel (PHP, React) applications with end-to-end—RESTful APIs, authentication/authorization, admin UIs, and workflows, serving **8,000+** daily users.
- Led development of a shipment-intelligence platform in Laravel (REST APIs, auth, queue workers, SQL) enabling shipper-receiver status sync, and audit trails; improved handoffs and cut communication downtime by **55%**.
- Built an LLM-assisted analytics pipeline with embeddings for event classification and daily PM summaries; added prompt guardrails, offline evaluation harnesses, and spend/latency dashboards to keep quality and costs predictable.
- Containerized Laravel apps and workers with Docker and deployed to AWS ECS behind LB with auto scaling, RDS (MySQL), and S3/CloudFront so that releases were safe and stable, resulting in **99.9%** weekly uptime.

Data Engineering Intern (Backend & Data) *AllProperty*

Jun 2023 – Aug 2023

- Cleaned and normalized **800,000+** property records using Python ETL with schema validation, and LangChain/RAG enrichment so that analytics, search, and exports; resulting in field completeness rising from **38%** → **95%**.
- Prototyped a local-LLM workflow to summarize free-text notes and flag inconsistent entries so that analysts could triage anomalies during weekly checks; resulting in **80%** less manual cleanup and review time cut from **2h** to **25m**.
- Built interactive investor demos in React backed by a live Node.js/REST API and PostgreSQL so that prospects could run property search, valuation, and portfolio reviews in real time, resulting in clearer value and onboarding **7** investors.

Full-Stack Developer Intern *Brands&People*

Jun 2022 – Sep 2022

- Rebuilt pages in Next.js/TypeScript with SSR/SSG, image optimization and code-splitting so that mobile loads stayed fast on poor networks, which meant Core Web Vitals passed, resulting in LCP **4.2s** → **1.8s** and Lighthouse **58** → **95**.
- Implemented an accessible, componentized UI using Tailwind, semantic HTML, and ARIA so that screen-reader/keyboard users could navigate all flows, resulting in Lighthouse accessibility **72** → **98** and tickets **-80%**.
- Added PWA features (service worker, offline routes) and Cloudflare CDN so that returning visitors saw instant navigation on flaky networks, which meant fewer drop-offs, resulting in repeat-visit latency **-60%** and session duration **+18%**.

Projects

Short-Form Content Automation | *Python, FFmpeg, Selenium, Whisper, scikit-learn, XGBoost*

Sep 2024

- Scaled an automated short-form video pipeline which produced and published content cross-platform daily, which meant consistent distribution and monetization, resulting in **1 Billion+** total views and recurring revenue.
- Trained an embedding-based ranker (scikit-learn/XGBoost with labels from historical views and a vector index) so that content candidates were scored for expected virality, resulting in higher median 24-hr views.
- Built an end-to-end pipeline—Reddit ingestion → transcription → TTS (11Labs) → caption alignment → render → upload—with job queues and logging so that publishing was fully automated, resulting in consistent daily output.

Intelligent Contract Renewal Calendar | *FastAPI, OpenRouter, Tesseract OCR, SQLite, Docker*

Aug 2025

- Architected a Renewal Calendar: a FastAPI backend orchestrated Tesseract OCR + GPT-4 extraction so that contracts auto-populated the calendar/ICS, which meant PDF → schedule, resulting in **10x** faster turnaround.
- Designed a confidence-driven review flow (extraction_confidence, candidate dates, computed notice_deadline) so that edge cases were shown in the UI, which meant fewer errors, resulting in correction rate **-65%** and zero missed notice windows.
- Implemented delivery/ops—ICS export + SMTP email, one-command Docker setup, OpenAPI docs, and file cleanup—so that deploys and handoffs stayed simple, which meant faster onboarding, resulting in time-to-first-setup **5 min**.

Skills

Python, C++, JavaScript/TypeScript, SQL, React, Node.js, Laravel, Flask, Docker, AWS, Google Cloud, Kubernetes; PyTorch, TensorFlow, TinyGrad, scikit-learn, XGBoost, Pandas, NumPy, Whisper, LangChain, FFmpeg